

NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY, GREATER NOIDA, G.B. NAGAR
(AN AUTONOMOUS INSTITUTE)



Affiliated to

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW



Evaluation Scheme & Syllabus

For

Bachelor of Technology

Computer Science and Engineering (Artificial Intelligence and Machine Learning)

Third Year

(Effective from the Session: 2025-26)

**NOIDA INSTITUTE OF ENGINEERING & TECHNOLOGY, GREATER NOIDA, GAUTAM BUDDH NAGAR
(AN AUTONOMOUS INSTITUTE)**

**Bachelor of Technology
Computer Science and Engineering (Artificial Intelligence and Machine Learning)**

**Evaluation Scheme
SEMESTER-VI**

Sl. No.	Subject Codes	Subject	Types of Subjects	Periods			Evaluation Schemes				End Semester		Total	Credit
				L	T	P	CT	TA	TOTAL	PS	TE	PE		
1	BCSAI0601	Natural Language Processing	Mandatory	3	1	0	30	20	50		100		150	4
2		Departmental Elective-III	Departmental Elective	3	0	0	30	20	50		100		150	3
3		Departmental Elective-IV	Departmental Elective	3	0	0	30	20	50		100		150	3
4		Open Elective I	Open Elective	3	0	0	30	20	50		100		150	3
5	BCSAI0651	Natural Language Processing Lab	Mandatory	0	0	2				25		25	50	1
6	BCSAI0652	Generative AI	Mandatory	0	0	6				50		100	150	3
7	BCSE0653	Software Engineering and Design	Mandatory	0	0	6				50		100	150	3
8	BCSE0659	Mini Project	Mandatory	0	0	6				50		100	150	3
9	BNC0601/ BNC0602	Constitution of India, Law and Engineering /Essence of Indian Traditional Knowledge	Compulsory Audit	2	0	0	30	20	50		50		100	NA
10		MOOCs (Essential for Hons. Degree)	MOOCs											
		TOTAL		14	1	20	120	80	200	175	400	325	1100	23

*** List of MOOCs Based Recommended Courses for Third year (Semester-VI) B. Tech Students**

S. No.	Subject Code	Course Name	University / Industry Partner Name	No of Hours	Credits
1	BMC0084	Introduction to ML and AI	Infosys Wingspan (Infosys Springboard)	62h 54m	4
2	BMC0087	JavaScript Essential	Infosys Wingspan (Infosys Springboard)	19h 42m	1.5
3	BMC0093	Natural Language Processing using Python	Infosys Wingspan (Infosys Springboard)	15h 45m	1

PLEASE NOTE: -

- **Compulsory Audit (CA) Courses (Non-Credit - BNC0601/BNC0602)**
 - All Compulsory Audit Courses (a qualifying exam) do not require any credit.
 - The total and obtained marks are not added to the grand total.

Abbreviation Used:

L: Lecture, T: Tutorial, P: Practical, CT: Class Test, TA: Teacher Assessment, PS: Practical Sessional, TE: Theory End Semester Exam.,
CE: Core Elective, OE: Open Elective, DE: Departmental Elective, PE: Practical End Semester Exam, CA: Compulsory Audit,
MOOCs: Massive Open Online Courses.

DEPARTMENTAL ELECTIVES

Sl. No.	Subject Codes	Subject Name	Types of Subjects	Bucket Name	Branch	Semester
1	BCSE0611	CRM Development	Departmental Elective-III	CRM-RPA	CSE (AIML)	6
2	BCSE0613	Robotics Process Automation (RPA)	Departmental Elective-IV		CSE (AIML)	6
3	BCSAI0617	Programming for Data Analytics	Departmental Elective-III	Data Analytics	CSE (AIML)	6
4	BCSAI0622	Social Media Analytics	Departmental Elective-IV		CSE (AIML)	6
5	BCSAI0612	Advanced Java Programming	Departmental Elective-III	Full Stack Development	CSE (AIML)	6
6	BCSE0614	Web Development using MEAN Stack	Departmental Elective-IV		CSE (AIML)	6
7	BCSAI0614	Development in Swift Explorations and Data Collections	Departmental Elective-III	Mobility Management	CSE (AIML)	6
8	BCSAI0620	Augmented Reality and Virtual Reality	Departmental Elective-IV		CSE (AIML)	6



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Course Code: BCSAI0601		Course Name: Natural Language Processing		L	T	P	C
Course Offered in: AI & AIML				0	0	6	3
Pre-requisite: Machine learning, formal languages, Data Structures, Algorithms Probability and Statistics							
Course Objectives: The goal of natural language processing (NLP) is to design and build computer systems that are able to analyze natural languages like English, and that generate their outputs in a natural language.							
Course Outcome: After completion of the course, the student will be able to					Bloom’s Knowledge Level (KL)		
CO1	Understand and explain the fundamentals of Natural Language Processing					K2	
CO2	Apply text vectorization and similarity techniques					K3	
CO3	Analyze text analytics methods					K4	
CO4	Demonstrate knowledge of sequential models and transformer architectures					K3	
CO5	Evaluate real-world NLP applications					K5	
Course Contents / Syllabus							
Module 1	Introduction to Natural Language Processing					14 hours	
Definition, Applications and emerging trends in NLP, Challenges. Ambiguity. NLP tasks using NLTK: Tokenization, stemming, lemmatization, stop-word removal, POS tagging, Parsing, Named Entity Recognition, coreference resolution.							
Module 2	Text Vectorization and Similarity					14 hours	
Text Vectorization: Bag-of-Words model, N-Gram Models and vector space models, Term Presence, TF-IDF Textual Similarity: Cosine similarity, Word Mover’s distance, Word embeddings: Word2Vec, GloVe.							
Module 3	Text Analytics					14 hours	
Text classification, Sentiment Analysis, Topic modelling, LSA, LDA, Opinion Mining, Information Extraction, Information Retrieval.							
Module 4	Sequential Modelling and Transformers					15 hours	
Sequential data, Introduction to sequence models - RNN and LSTM, Attention Mechanism, Transformer, Transformer-based models: BERT, FastText, Elmo, GPT, T5, Introduction to Hugging Face Transformers.							
Module 5	Applications and Case Studies					15 hours	
NLP applications: Machine translation: Rule-based and statistical approaches, Text summarization Dialog systems, conversational agents and chatbots. Automatic Document Separation: A Combination of Probabilistic Classification and Finite-State Sequence Modelling: Introduction, Related Work, Data Preparation, Document Separation as a Sequence Mapping Problem, Results.							
Total Lecture Hours					72 hours		
Textbook:							
S.No	Book Title					Author	
1.	Speech and Language Processing, 3rd Edition, Pearson Education, 2008					Daniel Jurafsky, James H.Martin	
2.	Deep Learning for Natural Language Processing, 1st Edition, Apress, 2018					Palash Goyal, Sumit Pandey, Karan Jain	
Reference Books:							
S.No	Book Title					Author	
1.	Natural Language Understanding, 2nd Edition, Pearson, 1995					James Allen	



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B. TECH THIRD YEAR (ELECTIVE III)

Course Code	BCSAI0612	L T P	Credits
Course Title	ADVANCED JAVA PROGRAMMING	3 0 0	3

Course objective:

Objective of this course is to provide the ability to design console based, GUI based ,web based applications, integrated development environment to create, debug and run multi-tier and enterprise-level applications.

Pre-requisites: Basics of C, C++, and basic concept of Core JAVA.

Course Contents / Syllabus

UNIT-I	Introduction	8 Hours
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JDBC: Introduction, JDBC Driver, DB Connectivity, Driver Manager, Connection, Statement, Result Set, Prepared Statement, Transaction Management, Stored Procedures.

Servlet: Servlet Overview, Servlet API, Servlet Interface, Generic Servlet, HTTP Servlet, Servlet Life Cycle, Redirect requests to other resources, Session Tracking, Event and Listener.

UNIT-II	JSP	8 Hours
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JSP: Introduction, Overview, JSP Scriptlet Tag, JSP expression Tag, JSP declaration Tag, Life Cycle of JSP, JSP API, Implicit Objects: JSP request, JSP response, JSP config, JSP session, JSP Application, JSP Page Context; JSP Page, JSP Exception.

UNIT-III	Spring 5.0	8 Hours
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Spring 5.0: Spring Core Introduction and Overview, Managing Beans, The Spring Container, The Factory Pattern, Dependency Injection (DI), Spring Managed Bean Lifecycle, Constructor Injection, Metadata/Configuration: Life Cycle Annotations, Java Configuration, XML Free configuration.

UNIT-IV	Spring MVC & Spring Boot	8 Hours
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Spring MVC: Introduction/Developing Web Application with Spring MVC, Advanced Techniques, Spring Controllers

Spring Boot: Spring Boot Starters, CLI, Application Class, Logging, Auto Configuration Classes, Spring Boot dependencies, Spring data JPA introduction and Overview.

UNIT-V	JPA	8 Hours
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JPA: Introduction & overview of data persistence, Overview of ORM tools, Understanding JPA, Entities: Requirement for Entity Class, Persistent Fields and Properties, Primary keys in Entries, Entity Management, Querying Entities, Entities Relationships.

Course outcome: After completion of this course students will be able to

CO 1	Understand the concept of implementing the connection between Java and Database using JDBC.	K2, K4
CO 2	Understand, Analyse, and Build dynamic web pages for server-side programming	K2, K3
CO 3	Analyze and design the Spring Core Modules and DI to configure and wire beans (application objects) together	K4,K5



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B. TECH THIRD YEAR (ELECTIVE IV)

Course Code	BCSE0614	L T P	Credits
Course Title	WEB DEVELOPMENT USING MEAN STACK	3 0 0	3

Course objective:

This course focuses on how to design and build static as well as dynamic webpages and interactive web applications. Students examine advanced topics like Angular, nodejs, Mongodb and Express framework for interactive web applications that use rich user interfaces.

Pre-requisites: Basic knowledge of HTML, CSS and ES6 required.

Course Contents / Syllabus

UNIT-I	Introduction to Nodejs	8 Hours
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Installing Nodejs, Node in-built packages (buffer, fs, http, os, path, util, url) Node.js modules, File System Module, Json data, Http Server and Client, Error handling with appropriate HTTP, Callback function, asynchronous programming REST API's(GET, POST PUT, DELETE UPDATE), GraphQL, Promises, Promise Chaining, Introduction to template engine (EJS).

UNIT-II	Express Framework	8 Hours
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Configuring Express, Postman configuration, Environment Variables, Routing, Defining pug templates, HTTP method of Express, URL binding, middleware function, Serving static files, Express sessions, REST full API's, FORM data in Express, document modeling with Mongoose.

UNIT-III	Basics of Angular js	8 Hours
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Typescript, Setup and installation, Power of Types, Functions, Function as types Optional and default parameters, Arrow functions, Function overloading, Access modifiers, Getters and setters, Read-only & static, Abstract classes, Interfaces, Extending and Implementing Interface, Import and Export modules.

UNIT-IV	Building Single Page App with Angular js	8 Hours
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MVC Architecture, One-way and Two-way data binding, AngularJS Expressions, AngularJS Controllers, AngularJS Modules, adding controller to a module, Component, Dependency Injection, Filters, Tables, AngularJS Forms and Forms validation, Select using ng-option, AngularJS AJAX.

UNIT-V	Connecting Angular js with MongoDB	8 Hours
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Environment Setup of Mongodb, data modeling, The current SQL/NoSQL landscape, Create collection in Mongodb, CRUD Operations in MongoDB. Mongo's feature set, Introduction to Mongoose, understanding mongoose schemas and datatypes, Connecting Angular with mongoDB using API.

Course outcome: After completion of this course students will be able to

CO 1	Explain, analyze and apply the role of server-side scripting language like Nodejs in the workings of the web and web applications.	K2, K3
CO 2	Demonstrate web application framework i.e., Express is to design and implement typical dynamic web pages and interactive web based applications.	K3, K6



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B. TECH THIRD YEAR			
Course Code	BCSAI0651	L T P	Credit
Course Title	Natural Language Processing Lab	0 0 4	2
List of Experiments			
Sr. No.	Name of Experiment	CO	
1.	1. Introduction to NLP - Tokenization of Sentences and Words using NLTK and spaCy - Stemming and Lemmatization on sample text - Stop-word Removal from a document - Part-of-Speech (POS) Tagging of a given sentence - Parsing and Chunking using regex and spaCy - Named Entity Recognition (NER) using spaCy - Coreference Resolution using neuralcoref (or similar)		
2.	2. Text Vectorization and Similarity - Bag-of-Words (BoW) vectorization and representation - TF-IDF Implementation and comparison with BoW - N-Gram Model (uni-, bi-, tri-gram) generation from corpus - Cosine Similarity computation between text documents - Word2Vec word embeddings using gensim on a custom corpus - GloVe Embeddings loading and vector representation - Text Similarity using Word Mover's Distance (WMD)		
3.	3. Text Analytics - Text Classification using Naïve Bayes/SVM with TF-IDF - Sentiment Analysis using TextBlob and VADER - Topic Modeling using Latent Dirichlet Allocation (LDA) - Topic Modeling using Latent Semantic Analysis (LSA) - Opinion Mining on product/service reviews dataset - Information Extraction (IE) from structured/unstructured documents - Information Retrieval system with ranking using TF-IDF		
4.	4. Sequential Modeling and Transformers - Sequence Classification using RNN/LSTM (Keras/TensorFlow) - Implementing Attention Mechanism (Basic custom model) - Fine-Tuning BERT for text classification using Hugging Face - Sentence Embeddings using BERT and GPT-2 - Text Generation using GPT-2 on a custom dataset - Named Entity Recognition using Transformers (BERT)		
5.	5. Applications and Case Studies - Machine Translation using MarianMT or T5 (Hugging Face) - Text Summarization using BART or T5 - Chatbot Development using pre-trained transformer (DialoGPT/ChatGPT-style) - Automatic Document Classification and Separation using ML		

 NIET Greater Noida Autonomous Institute	NOIDA INSTITUTE OF ENGINEERING AND TECHNOLOGY GREATER NOIDA-201306 (An Autonomous Institute) School of Computer Science in Emerging Technologies	
	Hybrid Probabilistic + FSM-Based Sequence Modeling for document boundary detection	

Course Code: BCSAI0652		Course Name: Generative AI		L	T	P	C
Course Offered in: CSE(AI)/CSE(AIML)				0	0	6	3
Pre-requisite: Statistics, Machine Learning, Deep Learning							
Course Objectives: To equip students with the knowledge and skills to build and fine-tune generative AI models—including autoencoders, GANs, transformers, and diffusion models—and apply them ethically in real-world domains such as image synthesis, text generation, and healthcare.							
Course Outcome: After completion of the course, the student will be able to				Bloom’s Knowledge Level (KL)			
CO1	Apply the concepts of discriminative vs. generative models.			K3			
CO2	Describe GMMs and HMMs via Expectation-Maximization.			K2			
CO3	Analyze VAEs with a focus on the reparameterization trick.			K4			
CO4	Design GANs for realistic image generation and training challenges.			K3			
CO5	Develop Autoencoders for dimensionality reduction and image reconstruction.			K4			
Course Contents / Syllabus							
Module 1		Foundations of Generative Model				14 hours	
Introduction to Generative vs. Discriminative Models, Probabilistic Foundations: Joint, Marginal, and Conditional Distributions, Maximum Likelihood Estimation (MLE) and KL Divergence, Latent Variable Models: GMM, HMM, Use Cases of Generative AI: Image synthesis, Text generation, Healthcare							
Module 2		Classical Models & Autoencoders				14 hours	
Gaussian Mixture Models (GMM), Expectation-Maximization, Hidden Markov Models (HMMs) Autoencoders (AE): Undercomplete, Denoising, Sparse Autoencoders, Variational Autoencoders (VAE): ELBO, Reparameterization Trick, Applications: Image reconstruction, Denoising, Latent Space Interpolation.							
Module 3		Generative Adversarial Networks (GANs)				14 hours	
GAN Architecture: Generator and Discriminator, Training Challenges: Mode Collapse, Vanishing Gradients, GAN Variants: DCGAN, cGAN, CycleGAN, StyleGAN2, BigGAN, Evaluation Metrics: FID, IS, PR Curves, Applications: Image synthesis, Super-resolution, Image translation, etc.							
Module 4		Attention Mechanism, Transformers & Large Language Models				15 hours	
Self-Attention, RNN vs. Attention, Multi-Head Attention, Transformer Architecture: Encoder-Decoder, Positional Encoding, Generative Transformers: GPT (Generative Pretrained Transformer), BERT (Masked Language Model), Fine-Tuning Pretrained Models: GPT-3, T5, BERT, and specialized applications like Text Summarization, Question Answering, LLMs in Action: GPT-3, GPT-4, LLaMA, Prompt Engineering: Understanding and crafting prompts to guide LLM outputs, Applications: Text Generation, Code generation, Conversational AI							
Module 5		Advanced Generative Models & Fine-Tuning				15 hours	
Diffusion Models: Forward and Reverse Process, Denoising Diffusion Probabilistic Models (DDPM) Flow-Based Models: NICE, RealNVP, Glow, Fine-Tuning Pretrained Models: GPT-2, Stable Diffusion Multimodal Models: CLIP, DALL·E, Flamingo, GPT-4 + Multimodal Inputs, Ethical Issues: Deepfakes, AI Bias, Copyright, Content Authenticity, and Responsible AI							
Total Lecture Hours						72 hours	
Textbook:							
S. No	Book Title			Author			
1.	Probabilistic Machine Learning: From Algorithms to Applications			Kevin P. Murphy			

2.	Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play	David Foster	
3.	Hands-On Generative AI with Transformers and Diffusion Models	Omar Sanseviero	
Reference Book:			
S. No	Book Title	Author	
1.	Machine Learning: From the Classics to Deep Networks, Transformers, and Diffusion Models.	Sergios Theodoridis	
2.	Essentials of Generative AI	Takeshi Okadome (Springer)	
NPTEL/ YouTube/ Faculty Video Link:			
Module 1	https://www.youtube.com/watch?v=hjsZSmL67Ck		
Module 2	https://www.youtube.com/watch?v=rVfZHWTwXSA		
Module 3	https://www.youtube.com/watch?v=qJeaCHQ1k2w		
Module 4	https://www.youtube.com/watch?v=AALBGpLbj6Q		
Module 5	https://www.youtube.com/watch?v=M1pF0QGDQyU		
Mode of Evaluation			
CIE		ESE	Total
PS			
50		100	150

List Of Practical's (Indicative & Not Limited To)		
S.NO.	PRACTICAL (Suggestive List of Practical)	CO
1.	Implement and visualize Joint, Marginal, and Conditional Distributions using synthetic data	CO1
2.	Perform Maximum Likelihood Estimation and compute KL Divergence for simple distributions	CO1
3.	Build and evaluate a Gaussian Mixture Model (GMM) using Expectation-Maximization on real-world data	CO1
4.	Implement a simple Hidden Markov Model (HMM) for weather prediction or POS tagging	CO1
5.	Implement an Undercomplete Autoencoder for image compression using MNIST	CO1
6.	Build and compare Denoising and Sparse Autoencoders for noisy image reconstruction	CO1
7.	Implement a Variational Autoencoder (VAE) and visualize latent space interpolation	CO2
8.	Implement a basic GAN for image generation using synthetic data	CO2
9.	Build and compare DCGAN and cGAN on image dataset (e.g., CelebA or FashionMNIST)	CO2
10.	Implement and demonstrate CycleGAN for style transfer	CO2
11.	Evaluate GAN performance using FID and Inception Score	CO2
12.	Implement self-attention and multi-head attention mechanisms	CO2
13.	Build a transformer encoder-decoder model for text summarization	CO2
14.	Use a pre-trained GPT model (e.g., GPT-2 or GPT-3) for text generation and prompt engineering	CO3
15.	Fine-tune a pre-trained BERT model for question answering or sentiment analysis	CO3
16.	Implement and visualize denoising diffusion probabilistic models (DDPM) for image synthesis	CO3
17.	Build a simple Flow-Based generative model using RealNVP or Glow	CO3
18.	Use CLIP or DALL·E to generate images from text prompts	CO3

19.	Demonstrate multimodal capabilities using Flamingo or GPT-4 with image and text inputs	CO3
20.	Case study: Analyze ethical concerns in generative AI with examples (deepfakes, bias, and responsible AI frameworks)	CO3



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Course Code	: BCSE0653	L T P	Credits
Course Title	Software Engineering & Design	[0-0-6]	3
Course objective: To help students understand all phases of the Software Development Life Cycle (SDLC) both theoretically and practically, enabling them to systematically apply principles of analysis, design, development, testing, and maintenance to build cost-effective software solutions and become competent software engineering professionals			
Pre-requisites: Basic knowledge of computer fundamentals and software development processes			
Course Contents / Syllabus			
UNIT-I	Introduction and development models	16 Hours	
Evolving role of software, Software Characteristics, Software crisis, silver bullet, Software myths, Software Engineering Phases, Team Software Process (TSP), Emergence of software engineering, Software process, project and product, Software Process Models: Waterfall Model, Prototype Model, Spiral Model, Iterative Model, Incremental Model, Agile Methodology: Scrum Sprint, Scrum Team, Scrum Master, Product Owner, Kanban framework.			
UNIT-II	Software Requirement Quality Assurance	16 Hours	
Software Requirement Specifications (SRS): Requirement Engineering Process: Elicitation, Analysis, Documentation, Review and Management of User Needs, Feasibility Study, Information Modelling, Use Case Diagram, Data Flow Diagrams, Entity Relationship Diagrams, Decision Tables, SRS Document, IEEE Standards for SRS. Software Quality Assurance (SQA): Quality concepts, SQA activities, Formal approaches to SQA; Statistical software quality assurance; CMM, The ISO standard.			
UNIT-III	Software Design	16 Hours	
Design principles, the design process; Design concepts: refinement, modularity: Cohesion, Coupling, Effective modular design: Functional independence, Design Heuristics for effective modularity, Software architecture: Function Oriented Design, Object Oriented Design: OOPs concepts-Abstraction, object, classification, inheritance, encapsulation, UML Diagrams-Class Diagram, Interaction diagram, Activity Diagram, Control hierarchy: Top-Down and Bottom-Up Design, structural partitioning, software procedure			
UNIT-IV	Software Testing	16 Hours	
Software Testing: Testing Objectives, 7 Principals of Testing, Levels of Testing: Unit Testing, System Testing, Integration Testing, User Acceptance Testing, Regression Testing, Testing for Functionality and Testing for Performance, Top Down and Bottom-Up Testing Strategies: Test Drivers and Test Stubs, Structural (White Box Testing Testing), Functional Testing (Black Box Testing), Test Data Suit Preparation, Alpha, and Beta Testing of Products. Static Testing Strategies: Formal Technical Reviews (Peer Reviews), Walk Through, Code Inspection, Compliance with Design and Coding Standards, Test Management, Test Planning and Estimation, Test Monitoring and Control, Configuration Management, Risks and Testing, Defect Management, Tool Support for Testing, Effective Use of Tools.			
UNIT-V	Project Maintenance Management Concepts	16 Hours	
Software Maintenance: Preventive, Corrective and Perfective Maintenance, Cost of Maintenance, Need for Maintenance. Project management concepts, Planning the software project, Estimation: Software Measurement and Metrics, Various Size Oriented Measures-LOC based, FP based, Halestead’s Software Science, Cyclomatic Complexity Measures: Control Flow Graphs, Use-case based empirical estimation COCOMO- A Heuristic estimation technique, staffing level estimation, team structures, risk analysis and management. Configuration Management, Software reengineering, reverse engineering, restructuring forward engineering, Clean Room software engineering. Case Tools			
Course outcome: After completion of this course students will be able to			

CO 1	Understand various software characteristics and analyze different software Development Models	K4
CO2	Demonstrate the concept of SRS and apply basic software quality assurance practices.	K3
CO3	Understand design principles and logic to effectively compare various software design methods.	K4
CO4	Apply various testing techniques.	K3
CO5	Maintain and apply software project management tools for software development.	K5

Text books:

1.	KK Aggarwal and Yogesh Singh, Software Engineering, New Age International Publishers 3RDEdition	
2.	RS Pressman, Software Engineering: A Practitioners Approach, McGraw Hill. 7thEdition	
3.	Rajib Mall, Fundamentals of Software Engineering, PHI Publication.4th Edition	

Links: NPTEL/You Tube/Web Link

Unit 1	https://www.youtube.com/watch?v=bLrbX4ZCQeY
Unit 2	https://www.youtube.com/watch?v=ZloPeQA1G4E
Unit 3	https://www.youtube.com/watch?v=rpk7fSkTIu8
Unit 4	https://www.youtube.com/watch?v=T0TynxN77oY
Unit 5	https://www.youtube.com/watch?v=nulFv99VBGs

List of Practical

Sr. No.	Program Title	CO Mapping
1	Team formation and allotment of Mini project: Problem statement, Literature survey, Requirement. analysis.	CO1
2	Draw the use case diagram	CO2
3	Draw the Data Flow Diagram (DFD): All levels.	CO2
4	Design an ER diagram for with multiplicity	CO2
5	Prepare SRS document in line with the IEEE recommended standards.	CO2
6	Draw State chart diagram.	CO3
7	Draw Object and Class diagram.	CO3
8	Create Interaction diagram: sequence diagram for SDD	CO3
9	Create Interaction diagram: collaboration diagram for SDD.	CO3
10	Create Activity diagram	CO3
11	Create Component diagram	CO3
12	Create Deployment diagram	CO3
13	Estimation of Test Coverage Metrics and Structural Complexity.	CO4
14	Design and develop a program in a language of your choice to solve the triangle problem defined	CO4

	as follows: Accept three integers which are supposed to be the three sides of a triangle and determine if the three values represent an equilateral triangle, isosceles triangle, scalene triangle, or they do not form a triangle at all. Assume that the upper limit for the size of any side is 10. Derive test cases for your program based on boundary-value analysis, execute the test cases, and discuss the results	
15	Design, develop, code, and run the program in any suitable language to solve the commission problem. Analyze it from the perspective of boundary value testing, derive different test cases, execute these test cases, and discuss the test results.	CO4
16	Design and develop a program in a language of your choice to solve the triangle problem defined as follows: Accept three integers which are supposed to be the three sides of a triangle and determine if the three values represent an equilateral triangle, isosceles triangle, scalene triangle, or they do not form a triangle at all. Assume that the upper limit for the size of any side is 10. Derive test cases for your program based on equivalence class partitioning, execute the test cases, and discuss the results.	CO4
17	Design and develop a program in a language of your choice to solve the triangle problem defined as follows: Accept three integers which are supposed to be the three sides of a triangle and determine if the three values represent an equilateral triangle, isosceles triangle, scalene triangle, or they do not form a triangle at all. Derive test cases for your program based on decision-table approach, execute the test cases, and discuss the results.	CO4
18	Create test cases for a program which determine whether an integer is prime or not by using path testing.	CO4
19	Create test cases for a program which determine whether an integer is prime or not by using Cyclomatic complexity.	CO4
20	Consider a program to input two numbers and print them in ascending order. Find all du paths and identify those du-paths that are not feasible. Also find all dc paths and generate the test cases for all paths (dc paths and non dc paths).	CO4
21	Consider the code to arrange the nos. in ascending order. Generate the test cases for loop coverage and path testing. Check the adequacy of the test cases through mutation testing and compute the mutation score for each.	CO4
22	Write Test cases for any Known Application (e.g., Banking Application)	CO4
23	Create a test plan document for any application (e.g., Library Management System)	CO4
24	Study of any testing tool (e.g., Win Runner)	CO4
25	Study of any bug tracking tool (e.g., Bugzilla, Bug bit)	CO4
26	Study of any test management tool (e.g., Test Director)	CO4
27	Study of any open source-Testing tool (e.g., Test link, Test Rail)	CO4
28	Study of any web testing tool (e.g., Selenium)	CO4
29	Mini Project with CASE tools.	CO5
30	Case Study Provided by Industry.	CO5

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B. TECH. THIRD YEAR 5th/ 6th					
Course code	BNC0601/BNC0602	L	T	P	Credits
Course Title	Constitution of India, Law and Engineering /Essence of Indian Traditional Knowledge	2	0	0	2
Course objective: This course aims to provide basic knowledge about different theories of society, state and polity in India, Indian literature, culture, Indian religion, philosophy, science, management, cultural heritage and different arts in India					
Pre-requisites: Computer Organization and Architecture					
Course Contents / Syllabus					
UNIT-I	SOCIETY STATE AND POLITY IN INDIA				8 Hours
State in Ancient India: Evolutionary Theory, Force Theory, Mystical Theory Contract Theory, Stages of State Formation in Ancient India, Kingship , Council of Ministers Administration Political Ideals in Ancient India Conditions’ of the Welfare of Societies, The Seven Limbs of the State, Society in Ancient India, Purusārtha, Varnāshrama System, Āshrama or the Stages of Life, Marriage, Understanding Gender as a social category, The representation of Women in Historical traditions, Challenges faced by Women.					
UNIT-II	INDIAN LITERATURE, CULTURE, TRADITION, AND PRACTICES				8 Hours
Evolution of script and languages in India: Harappan Script and Brahmi Script. The Vedas, the Upanishads, the Ramayana and the Mahabharata, Puranas, Buddhist And Jain Literature in Pali,Prakrit And Sanskrit, Sikh Literature, Kautilya’s Arthashastra, Famous Sanskrit Authors, Telugu Literature, Kannada Literature,Malayalam Literature ,Sangama Literature Northern Indian Languages & Literature, Persian And Urdu ,Hindi Literature					
UNIT-III	INDIAN RELIGION, PHILOSOPHY, AND PRACTICES				8 Hours
Pre-Vedic and Vedic Religion, Buddhism, Jainism, Six System Indian Philosophy, Shankaracharya, Various Philosophical Doctrines, Other Heterodox Sects, Bhakti Movement, Sufi movement, Socio religious reform movement of 19th century, Modern religious practices.					
UNIT-IV	SCIENCE, MANAGEMENT AND INDIAN KNOWLEDGE SYSTEM				8 Hours
Astronomy in India, Chemistry in India, Mathematics in India, Physics in India, Agriculture in India, Medicine in India, Metallurgy in India, Geography, Biology, Harappan Technologies, Water Management in India, Textile Technology in India ,Writing Technology in India Pyrotechnics in India Trade in Ancient India/,India’s Dominance up to Pre-colonial Times.					
UNIT-V	CULTURAL HERITAGE AND PERFORMING ARTS				8 Hours
Indian Architect, Engineering and Architecture in Ancient India, Sculptures, Pottery, Painting, Indian Handicraft, UNESCO’S List of World Heritage sites in India, Seals, coins, Puppetry, Dance, Music, Theatre, drama, Martial Arts Traditions, Fairs and Festivals, UNESCO’S List of Intangible Cultural Heritage, Calenders, Current developments in Arts and Cultural, Indian’s Cultural Contribution to the World. Indian Cinema.					
COURSE OUTCOMES: After completion of this course students will be able to					
CO 1	Understand the basics of past Indian politics and state polity.				K2
CO 2	Understand the Vedas, Upanishads, languages & literature of Indian society.				K2